

CoreTech Cybersecurity Internship Task

Introduction

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Penetration-Testing-Ethical-Hacking-Lab/

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Penetration Testing Methodology and Ethical Hacking Lab

Objective

This project demonstrates the basic penetration testing methodology in a controlled and legal laboratory environment using:

- Kali Linux
- VirtualBox
- Metasploitable 2 (or DVWA)
- Nmap
- Metasploit Framework

⚠ Disclaimer

All testing was performed inside an isolated virtual lab environment. No attacks were conducted against public or unauthorized systems.

Contents

- Penetration Testing Phases
- Virtual Lab Setup
- Metasploit Practice
- Vulnerability Assessment
- Professional Penetration Test Report

Tools Used

- VirtualBox
- Kali Linux
- Metasploitable 2
- Nmap
- Metasploit Framework

Penetration Testing Phases

1. Reconnaissance

Reconnaissance is the information gathering phase.

Activities include:

- Finding IP addresses
- DNS enumeration
- WHOIS lookup
- Identifying operating systems
- Collecting publicly available information

Purpose:

Gather as much information as possible before testing.

2. Scanning

Scanning identifies live hosts and open services.

Common tools:

- Nmap
- Netdiscover

Example:

```
nmap -sV 192.168.56.101
```

Purpose:

- Open ports
- Running services
- Software versions

3. Exploitation

Attempt to exploit identified vulnerabilities.

Common tool:

Metasploit Framework

Example:

use exploit/unix/ftp/vsftpd_234_backdoor

Purpose:

Gain controlled access to the vulnerable system.

4. Post-Exploitation

Performed after successful access.

Activities:

- Privilege checking
- System information gathering
- User enumeration
- Maintaining access (only in authorized labs)

Purpose:

Determine business impact.

5. Reporting

The most important phase.

Includes:

- Executive Summary
- Findings
- Severity Levels
- Risk Analysis
- Recommendations

Purpose:

Help the organization fix security weaknesses.

Virtual Penetration Testing Lab

Software

- Oracle VirtualBox
- Kali Linux VM
- Metasploitable 2 VM

Network Configuration

Both virtual machines should use:

Host-Only Adapter

Example:

Kali Linux:
192.168.56.102

Metasploitable:
192.168.56.101

Verify connectivity:

ping 192.168.56.101

Expected:

Packets transmitted successfully.

Lab Diagram

```
graph TD
    KaliLinux[Kali Linux] ---| | VirtualBox[VirtualBox Host-Only Network]
    VirtualBox ---| | Metasploitable2[Metasploitable 2]
```

Virtual Penetration Testing Lab

Software

- Oracle VirtualBox
- Kali Linux VM
- Metasploitable 2 VM

Network Configuration

Both virtual machines should use:

Host-Only Adapter

Example:

Kali Linux:
192.168.56.102

Metasploitable:
192.168.56.101

Verify connectivity:

ping 192.168.56.101

Expected:

Packets transmitted successfully.

Lab Diagram

Kali Linux

|

VirtualBox Host-Only Network

|

Metasploitable 2

03_Metasploit_Basic_Commands.md

Basic Metasploit Commands

Start Metasploit

msfconsole

Search for an exploit

search vsftpd

Use exploit

use exploit/unix/ftp/vsftpd_234_backdoor

Show options

show options

Set target

set RHOSTS 192.168.56.101

Run exploit

run

View active sessions

sessions

Interact with session

sessions -i 1

Background session

background

Exit

exit

04_Vulnerability_Assessment.md

Basic Vulnerability Assessment

Target

Metasploitable 2

IP Address

192.168.56.101

Nmap Scan

Command

nmap -sV -A 192.168.56.101

Sample Findings

Port 21

FTP

vsftpd 2.3.4

Risk:

High

Port 22

SSH

OpenSSH

Risk:

Medium

Port 80

Apache HTTP

Risk:

Medium

Port 3306

MySQL

Risk:
Medium

Recommendations

- Update outdated software
- Disable unused services
- Restrict remote access
- Apply latest security patches
- Use strong authentication

05_Penetration_Test_Report.md

Professional Penetration Testing Report

Executive Summary

A controlled penetration test was performed on the Metasploitable virtual machine inside an isolated lab environment.

The objective was to identify security vulnerabilities and provide remediation recommendations.

Scope

Target:

Metasploitable 2

Testing Environment:

VirtualBox Host-Only Network

Tester:

Rubab Faiz

Methodology

1. Reconnaissance
2. Scanning
3. Vulnerability Identification
4. Exploitation Demonstration
5. Reporting

Findings

Finding 1

Service

FTP

Port

21

Severity

High

Description

Outdated vsftpd version detected.

Recommendation

Upgrade FTP server.

Finding 2

Service

Apache HTTP

Port

80

Severity

Medium

Recommendation

Update Apache and disable unnecessary modules.

Finding 3

Service

MySQL

Port

3306

Severity

Medium

Recommendation

Restrict remote database access.

Risk Summary

High Risk: 1

Medium Risk: 2

Low Risk: 0

Conclusion

The assessment identified several intentionally vulnerable services suitable for penetration testing practice.

All testing was performed inside a legal virtual laboratory.

No unauthorized systems were accessed.

Commands Used During the Lab

ip addr

ping 192.168.56.101

nmap -sV 192.168.56.101

nmap -A 192.168.56.101

msfconsole

search vsftpd

use exploit/unix/ftp/vsftpd_234_backdoor

show options

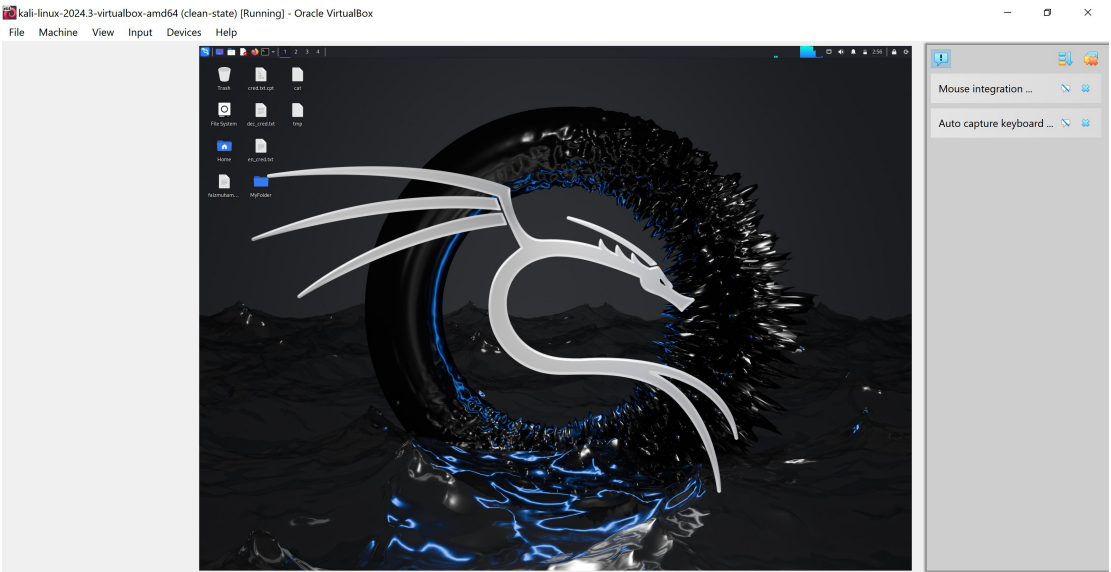
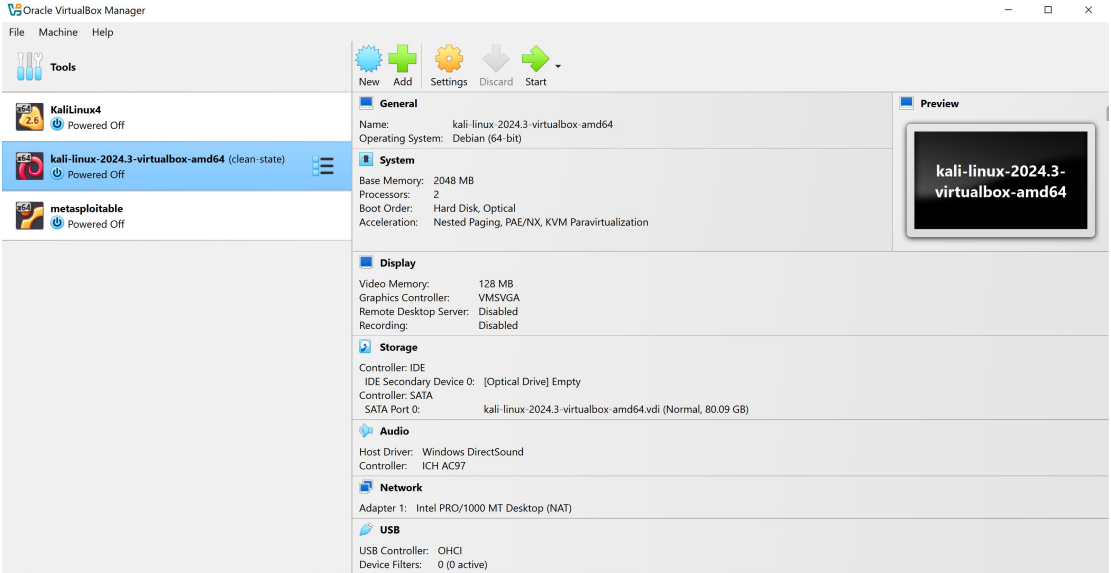
set RHOSTS 192.168.56.101

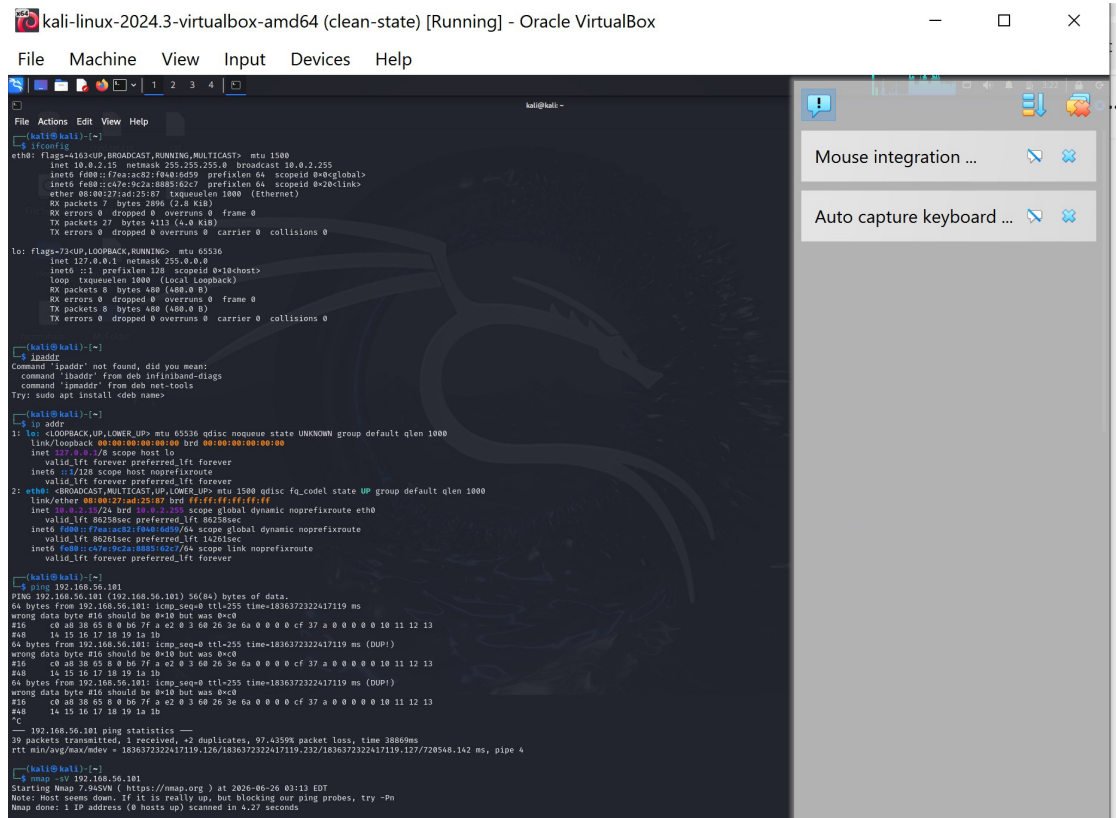
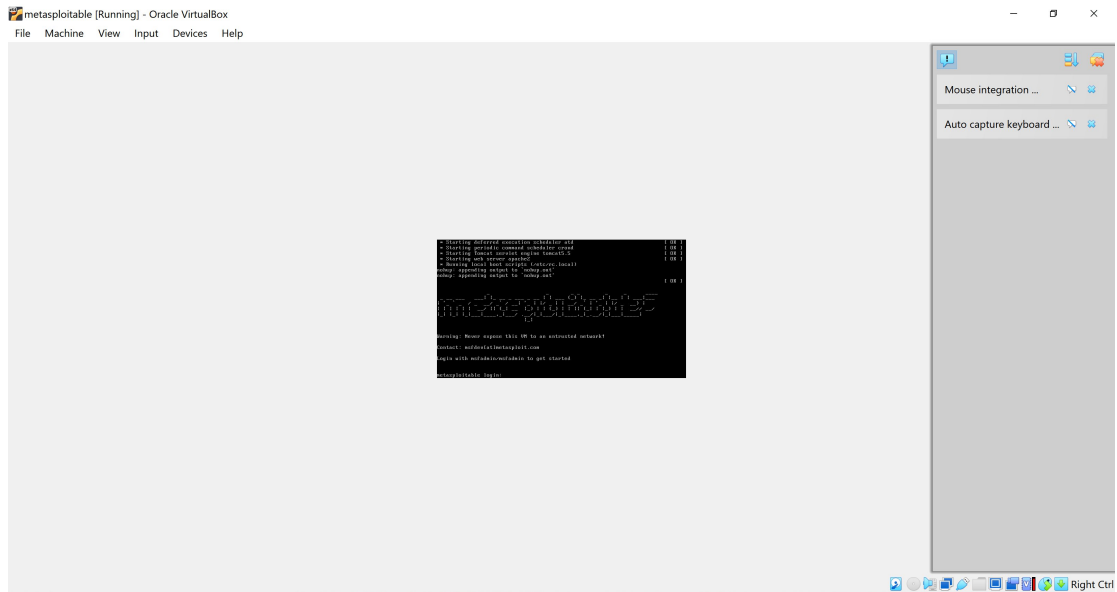
run

sessions

exit

Screenshots





Mouse integration ...  

Auto capture keyboard ...  